DISPENSARY MODULE (STUDENT HEALTH & MEDICAL ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Medical Staff • IT Team • Teachers • Parents
Related Modules: Student Management • Attendance • Academics • Notifications • Parent Portal • Analytics

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1. MODULE OVERVIEW

1.1 Purpose

The Dispensary Module manages all student health records, treatments, and medical resources within the school. It enables:

- Recording student health visits and treatments
- Managing medication inventory and usage
- Tracking referrals to external healthcare facilities
- Monitoring health trends and common conditions
- Ensuring timely communication with parents
- Supporting compliance with health and data privacy standards

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2. CORE OBJECTIVES

1. Maintain comprehensive student health profiles
2. Track all dispensary visits and treatments
3. Manage medication inventory and stock levels
4. Enable referrals and emergency response tracking
5. Provide analytics on student health trends
6. Integrate health data with attendance and academic performance
7. Ensure secure handling of sensitive medical information
8. Support real-time communication with parents and staff

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Student Falls Ill → Visits Dispensary → Diagnosis Recorded → Treatment Administered → Medication Updated → Parent Notified → Follow-Up/Referral → Records Stored

3.2 Daily Operational Flow

Student visits dispensary
↓
Medical staff logs visit and symptoms
↓
Treatment or medication administered
↓
Stock updated automatically
↓
Parent notified (if required)
↓
Referral issued (if serious case)
↓
Health records and analytics updated

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4. MODULE ARCHITECTURE

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Dispensary Module
│
├── Student Health Profiles
├── Visit & Treatment Records
├── Medication & Inventory Management
├── Referral Management
├── Emergency & Incident Handling
├── Analytics & Reporting
├── Notifications & Parent Communication
├── Staff & Resource Management
├── Integration Layer (Attendance/Academics)
├── Security & Compliance
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5. STUDENT HEALTH PROFILES

5.1 Features

- Maintain detailed health records per student

- Record allergies, chronic conditions, and medical history
- Track past visits and treatments

5.2 Data Model

HealthProfile

- profile_id
- student_id
- allergies
- chronic_conditions
- medical_history
- emergency_contact

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6. VISIT & TREATMENT RECORDS

6.1 Features

- Log each dispensary visit
- Record symptoms, diagnosis, and treatment
- Track attending medical staff
- Maintain historical records

6.2 Data Model

Visit

- visit_id
- student_id
- date
- symptoms
- diagnosis
- treatment
- staff_id
- notes

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7. MEDICATION & INVENTORY MANAGEMENT

7.1 Features

- Track medication stock levels

- Monitor expiry dates and batch numbers
- Automatic low-stock alerts
- Record medication usage per treatment

7.2 Data Model

Medication

- medication_id
- name
- quantity
- expiry_date
- batch_number
- reorder_level

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8. REFERRAL MANAGEMENT

8.1 Features

- Record referrals to hospitals or specialists
- Track referral status and follow-ups
- Maintain history of external treatments

8.2 Data Model

Referral

- referral_id
- student_id
- hospital_name
- reason
- date
- status

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9. EMERGENCY & INCIDENT HANDLING

9.1 Features

- Log critical incidents (injuries, severe illness)
- Emergency contact notifications
- First aid protocols and guidelines

- Track response time and actions taken

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10. ANALYTICS & REPORTING

10.1 Features

- Daily, monthly, and term health reports
- Common illness trends (e.g., flu, malaria)
- Visit frequency and absenteeism correlation
- Medication usage reports

10.2 Dashboards

- Health activity overview
- Condition trend analysis
- Referral statistics

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11. NOTIFICATIONS & PARENT COMMUNICATION

11.1 Features

- Alerts when a student visits the dispensary
- Notifications for medication administration
- Referral alerts
- Consent tracking for treatments

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12. STAFF & RESOURCE MANAGEMENT

12.1 Features

- Track medical staff schedules and workload
- Manage non-medical resources (equipment)
- Assign responsibilities for shifts

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13. INTEGRATION LAYER

13.1 Features

- Link health visits with attendance records
- Correlate health data with academic performance
- Sync with Student Management Module

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14. SECURITY & COMPLIANCE

14.1 Features

- Role-based access control for sensitive data
- Data encryption and secure storage
- Audit logs for all medical actions

14.2 Compliance

- Adherence to local health data protection laws
- Secure handling of minors' medical information

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15. MOBILE-FIRST DESIGN

15.1 Requirements

- Mobile-friendly interface for medical staff
- Quick data entry during emergencies
- Offline capability with sync

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16. SCALABILITY

- Support large student populations
- Multi-campus deployment
- Efficient handling of health data and reports

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17. IMPLEMENTATION GAPS (CRITICAL FIXES)

17.1 Current Issues Identified

- No detailed student health records
- Limited visit and treatment tracking
- Missing expiry and batch tracking for medications
- No advanced analytics or reporting
- Weak integration with attendance and academics
- Limited parent communication features

17.2 Required Fixes

- Implement full student health profiles
- Enhance visit and treatment logging
- Add medication expiry and batch tracking
- Build analytics dashboards
- Integrate with attendance and academic modules
- Enable automated parent notifications

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18. FUTURE ENHANCEMENTS

- Telemedicine integration
- AI-based health risk prediction
- Wearable device integration (health monitoring)
- Digital prescriptions and pharmacy integration
- Advanced epidemic tracking and alerts